Homework Assignment 2

Q1 and Q2, You can find solutions directly from Lec02 and Lec03 PPT.

Problem 1. (15 points) What is the output of the following queries.

An example of how to answer this question:

```
select distinct name
from    instructor
order by name  desc
```

Answer: List in alphabetic descending order the names of all instructors.

(a) (5 points)

```
select name
from instructor
 where salary > some (select salary
                      from instructor
                      where dept name = 'Biology');
```

(b) (10 points)

```
select distinct S.ID, S.name
from student as S
where not exists ( (select course_id
                    from course
                    where dept_name = 'Biology')
                   except
                    (select T.course_id
                     from takes as T
                     where S.ID = T.ID));
```

For (b), **also explain** what is the output of the **first nested query**, what is the output of the **second nested query**, and what are **"except" and "not exists".**

Problem 2. (10 points) Is the following SQL query Correct? If not, why? And how to fix it.

```
select dept_name, ID, avg (salary)
from instructor
group by dept_name;
```

Problem 3. (25 points) Consider the following relational database where the primary keys are underlined:

student(<u>student_id</u>, name)

course(<u>course_id</u>, classroom)

take(<u>student_id, course_id</u>)

(a) (10 points) Give an SQL data definition of the above database (i.e., write SQL commands that create tables). Integrity constraints (primary key and foreign keys) must be included in the data definition.

(b) (10 points) Write an SQL query that finds the IDs of the courses that John Smith has taken. (Hint: You need tables: student and take.)

(c) (5 points) Write an SQL query that finds the classrooms of the courses that John Smith has taken. (Hint: You need all three tables.)

Problem 4. (25 points) Consider the following relational database where the primary keys are underlined:

employee(<u>id</u>, name, team name, annual salary)

project(<u>project_name</u>, budget)

team(<u>team_name</u>, manager_id) // manager id is a foreign key to the employee table

works_on(<u>team_name, project_name</u>)

Express each of the following queries in SQL.

(a) (5 points) Find the total amount of budget to support projects contributed by team 'Y'. (Hint: You need two table, and an aggregate function.)

(b) (10 points) Find the names of employees who earn more than any (all) participant in project 'Z'. (Hint: You may want to use a subquery and "all" operation.)

(c) (10 points) Find the names of the teams that do not participate in any project. (Hint: You may want to use "not in" operation and a subquery.)

Problem 5. (10 points) You can finish this question using our Lab environment and PostgreSQL.

Check the Appendix if you forget how to run PostgreSQL and the pokemon database in our csci366 server.

(a) (5 points) Write an SQL query to find type_1 and the average speed (rename it as "avg_speed") of each type_1 whose average speed is greater than 70. (Hint: You should use an aggregate function , "group by" and "having". You should ignore type_2.). Your output should be:

```
  type_1   |       avg_speed
----------+---------------------
 Dragon    |  83.0312500000000000
 Electric  |  84.5000000000000000
 Normal    |  71.5510204081632653
 Psychic   |  81.4912280701754386
 Fire      |  74.4423076923076923
 Flying    | 102.5000000000000000
 Dark      |  76.1612903225806452
(7 rows)
```

(b) (5 points) Write an SQL statement to find the number of distinct pokemons whose speed is greater than 145.

The result should be:

```
 count
-------
     6
(1 row)
```

Explain the difference between your solution to 5(b) and the following query:

select count(name), speed
from pokemon
where speed >145
group by speed;

Problem 6. (15 points) Use recursive SQL to find all managers (names) of David in the following relation named **emp**.

EmployeeID	EmployeeName	ManagerID
1	John	5
2	Mark	8
3	Steve	8
4	Tom	3
5	Lara	8
6	Simon	2
7	David	4
8	Ben	NULL
9	Stacy	2
10	Sam	5

Appendix

To run PostgreSQL in our server.

1. Using your chosen SSH client, connect to csci366.millersville.edu. You have done so in the previous lab.
2. Go to the directory for this course. This directory should contain pokemon.csv file. E.g. cd 366
3. Run the PostgreSQL client. To do so, type

   psql pokemon_[your username]

4. Once the PostgreSQL is running, you can check if you still have the pokemon table with 800 rows by typing:

   Select * From pokemon;